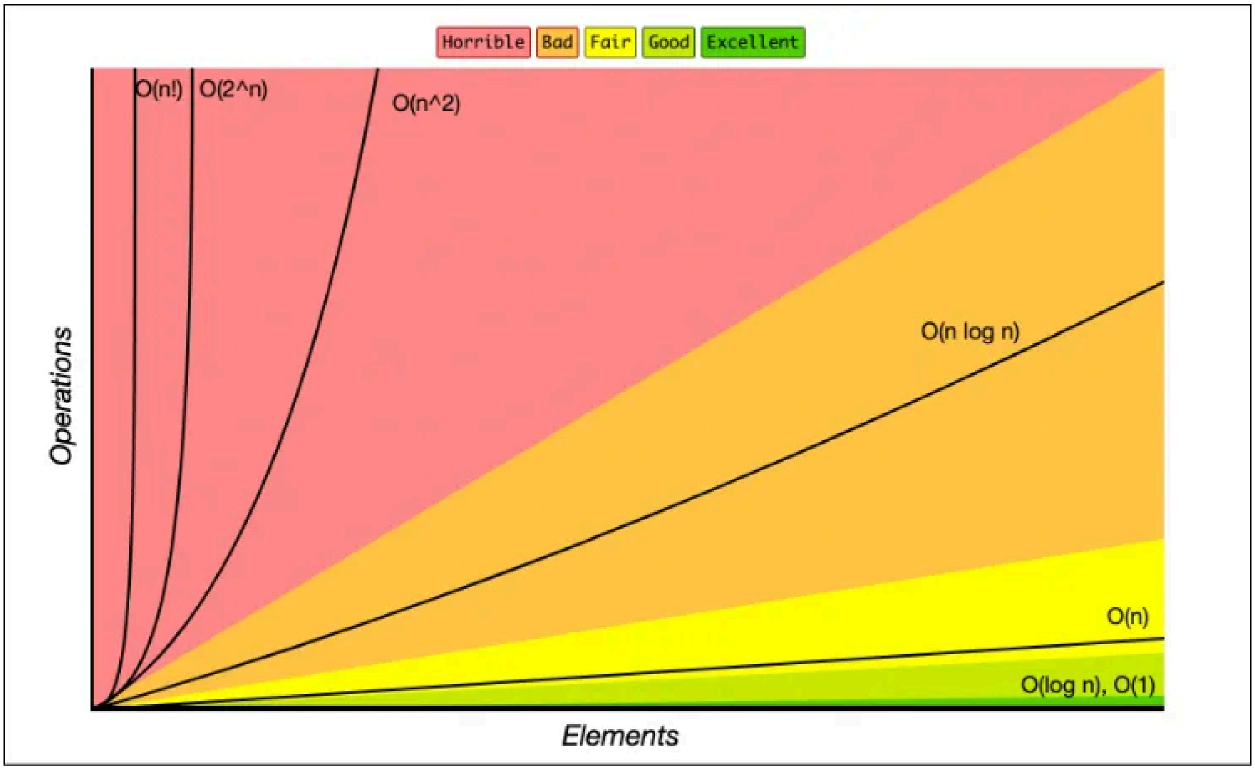
[Time Complexity Chart]():



Time Complexity MCQ

1. What will be the time complexity of the following code?

```
int i = 0;
while (i < 100) {
      i = i + 1;
      i = i * 2;
}
```

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: C. O(log n)

2. What will be the time complexity of the following code?

```
for (int i = 1; i <= n; i *= 2) {
  // Some operation
}
```

A. O(1)
B. O(n)
C. O(n/2)
D. O(log n)

Answer: D. O(log n)

3. What will be the time complexity of the following code?

```
int i = 0;
int j = 0;
while (i < n) {
   while (j < n) {
         j++;
   }
  i++;
}
```

A. O(1)
B. O(n)
C. O(n^2)
D. O(n^3)

Answer: C. O(n^2)

4. What will be the time complexity of the following code?
100*100

```
int sum = 0;
for (int i = 0; i < n; i++) {
   for (int j = i; j < n; j++) {
      sum += i * j;
   }
}
```

A. O(1)

B. O(n)
C. O(n^2)
D. O(n^3)

Answer: C. O(n^2)

5. What will be the time complexity of the following code?

```
int gcd(int a, int b) {
   if (b == 0) {
      return a;
   }
   return gcd(b, a % b);
}
```

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: C. O(log n)

6. What will be the time complexity of the following code?

```
double power(double base, int exponent) {
   if (exponent == 0) {
      return 1;
   }
   if (exponent < 0) {
      return 1 / (base * power(base, -exponent - 1));
   }
   return base * power(base, exponent - 1);
}
```

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: C. O(log n)

7. What will be the time complexity of the following code?

```
int arraySum(int arr[], int n) {
   if (n <= 0) {
      return 0;
   }
   return arr[n - 1] + arraySum(arr, n - 1);
}
```

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: B. O(n)

8. What will be the time complexity of the following code?

```
int fibonacci(int n) {
   if (n <= 1) {
      return n;
   }
   return fibonacci(n - 1) + fibonacci(n - 2);
}
```

A. O(log n)
B. O(n)
C. O(1^n)
D. O(2^n)

Answer: D. O(2^n)

9. What will be the time complexity of the following code?

```
int factorial(int n) {
   if (n <= 1) {
```

```
        return 1;
    }
    return n * factorial(n - 1);
}
```

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: B. O(n)

10. What will be the time complexity of the following code?

```
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        for (int k = 0; k < n; k++) {
            // Some operation
        }
    }
}
```

A. O(1)
B. O(n)
C. O(n^2)
D. O(n^3)

Answer: D. O(n^3)

11. What will be the time complexity of the following code?

```
if(i>j){
        j==0? j++ : j--;
}
```

A. O(1)
B. O(n)
C. O(n^2)
D. O(n log n)

Answer: A. O(1)

Space Complexity MCQ

Problem 12: Array Memory Usage

You have an array of n integers. What is the space complexity of storing this array in memory?

A. O(n)
B. O(1)
C. O(log n)
D. O(n^2)

Answer: A. O(n)

Problem 13: Recursive Function Memory

You have a recursive function that calls itself n times. What is the space complexity of the function call stack?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: C. O(log n)

Problem 14: Linked List Memory Usage

You have a singly linked list with n nodes. What is the space complexity of storing this linked list in memory?

A. O(n)
B. O(1)
C. O(log n)
D. O(n^2)

Answer: A. O(n)

Problem 15: Sorting Algorithm Space Complexity

You are using the merge sort algorithm to sort an array of n elements. What is the space complexity of merge sort?

A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)

Answer: B. O(n)

Problem 16: Space for a 2D Array

You have a 2D array of size m x n. What is the space complexity of storing this 2D array in memory?

A. O(m)
B. O(n)
C. O(m + n)
D. O(m * n)

Answer: D. O(m * n)

Problem 17: Space for a Binary Tree

You have a binary tree with n nodes. What is the space complexity of storing this binary tree in memory?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: B. O(n)

Problem 18: Space for a Hash Table

You are using a hash table to store n key-value pairs. What is the space complexity of the hash table?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: B. O(n)

Problem 19: Space for a Queue

You are implementing a queue data structure that can hold up to n elements. What is the space complexity of the queue?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: A. O(n)

Problem 20: Space for a Stack

You are implementing a stack data structure that can hold up to n elements. What is the space complexity of the stack?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: A. O(n)

Problem 21: Space for a String

You have a string of length n. What is the space complexity of storing this string in memory?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)

Answer: B. O(n)

Prepared By: [Abhishek Chauhan]